



Instituto Politécnico
de Castelo Branco



Instituto Politécnico de Castelo Branco
Escola Superior de Tecnologia

Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Internet Technologies**
Lecturer: Osvaldo Santos

Practical work on an introduction to Cloud Computing with Azure

Objectives of this sheet

Initial contact with the Azure public cloud computing platform

Bibliography and useful links

Azure doc official website

<https://docs.microsoft.com/en-us/azure/?product=popular>

Azure command reference in Powershell (CLI)

<https://learn.microsoft.com/en-us/cli/azure/>

Practical work on an introduction to Cloud Computing with Azure

In this assignment you will use your ipcbcampus student account to access Azure

At the end of the job, you must delete all Azure resources so that you can preserve your balance

1) Linux VM creation via the portal

- a) All the resources will be created in European Regions - for example Central France.
- b) Create a resource group called "AulaTI";
- c) In this resource group, create a VM with Ubuntu Server 20.04 LTS and Apache to implement a web server with the following characteristics:
 - i) 4 GB RAM, 2 vCPU, Standard SSD disc
 - ii) Authentication by username/password (instead of SSH public key)
 - iii) Open Internet ports: SSH, HTTP and HTTPS (22, 80 and 443)
 - iv) The Apache installation should be automatic
- d) Run the VM in Azure and test remote access via SSH
- e) Test the web server using a browser and the VM's public IP address

2) Creating a Windows VM via the portal

- a) In the previous resource group, create a VM with Windows Server 2022 Datacenter - Azure edition, to implement a web server, with the following characteristics:
 - i) 4 GB RAM, 2 vCPU
 - ii) Standard SSD disc
 - iii) RDP (Remote Desktop) access enabled and HTTP and HTTPS ports enabled.
- b) Run the VM in Azure and test remote access to the Windows Desktop via an RDP client application. On Windows, we recommend the "Remote Desktop" RDP client application, available in the Windows Store.



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3) Interaction with Azure via CLI - Command Line Interface

- a) If it's not installed, install the Azure Command Line Interface (CLI)

Tip: <https://learn.microsoft.com/en-us/cli/azure/install-azure-cli-windows?tabs=azure-cli>

- a) Open Microsoft Windows Powershell
- b) Log in to Azure using the "**az login**" command. Sign in with your ipcbcampus account
- c) Check which Ubuntu 20 images are available

Tip:

```
az vm image list --all --publisher Canonical --offer "Ubuntu" --sku 20
```

The image name is in the "urn" field.

- d) Create a script called **createUbuntu20.ps1**, which accepts a parameter for the region and creates an Ubuntu Server 20.04 LTS VM with the characteristics of point 1, in the region indicated.

Example: the command **"./createUbuntu20.ps1 franceCentral"** will create an Ubuntu Server 20 VM in the franceCentral region.

- e) Stop the VM, using the appropriate CLI command
- f) Delete the VM, using the appropriate CLI command
- g) Delete the resource group using the appropriate CLI command
- h) Log out of Azure using the "**az logout**" command.

4) Conclusion

- a) Does what you've just done in points 1, 2 and 3 fall under IaaS, PaaS or SaaS?

Now delete all the resources you have created